BROKERLENS

How the Business Price Is Estimated

A complete, printable breakdown of every input, formula, threshold, research source, output, and current limitation in the BrokerLens MVP.

CORE PRICING FORMULA

Normalized SDE x adjusted industry multiple
+ included inventory + excess assets - buyer-assumed debt

Current implementation review
Prepared July 22, 2026
Status: preliminary broker opinion methodology

Important: BrokerLens currently uses deterministic valuation rules. OpenAI research is displayed as supporting evidence but does not yet change the valuation.

Executive summary

BrokerLens currently calculates a preliminary business value from normalized seller earnings, a fixed industry starting multiple, six rule-based quality adjustments, and three transaction-balance adjustments. The result is presented as a low-to-high market range, a suggested asking price, and a likely sale range.

```
Value = SDE x adjusted multiple + inventory + excess assets - assumed debt
```

THE MOST IMPORTANT DISTINCTION

The OpenAI research feature is real and can search approved websites. However, its findings are not currently converted into pricing adjustments. The dollar estimate is produced entirely by the fixed calculation described in this document.

Contents

1. Inputs and how they are used

BrokerLens collects 20 business fields. Fifteen currently affect the dollar estimate directly. Other fields support AI research, the completion score, project identification, or future functionality.

Inputs that directly affect price

Input	Current use
Industry	Selects the fixed starting earnings multiple.
Net profit	Included in SDE.
Owner salary	Added back to SDE.
Interest	Added back to SDE.
Depreciation and amortization	Added back to SDE.
Verified one-time add-backs	Added back to SDE.
Revenue growth rate	Applies a fixed multiple adjustment.
Recurring revenue percentage	Applies a fixed multiple adjustment.
Owner dependence	Applies a fixed multiple adjustment.
Largest customer share	Applies a fixed multiple adjustment.
Years remaining on lease	Applies a fixed multiple adjustment.
Local growth	Applies a fixed multiple adjustment; entered manually.
Inventory included	Added dollar-for-dollar after applying the multiple.
Excess/non-operating assets	Added dollar-for-dollar.
Debt assumed	Subtracted dollar-for-dollar.

Inputs that do not directly affect price

Input	Current use
Business name	Identification, AI prompt, and completion score.
City and state	AI research location; city also affects completion score.
Business description	AI research context and completion score.
Approved domains	Restricts AI web search and affects completion score.

2. Seller's Discretionary Earnings

The first calculation normalizes earnings into Seller's Discretionary Earnings (SDE). This is intended to approximate the total economic benefit available to one owner-operator before financing and selected non-cash or non-recurring costs.

```
SDE = net profit + owner salary + interest + depreciation/amortization + one-time add-backs
```

Component	Why it is included	Current verification
Net profit	Starting accounting earnings figure.	User-entered.
Owner salary	Restores owner compensation to economic benefit.	User-entered.
Interest	Separates financing structure from operations.	User-entered.
Depreciation/amortization	Adds back non-cash accounting expenses.	User-entered.
One-time add-backs	Removes claimed non-recurring expenses.	User-entered; not verified.

CURRENT GUARDRAIL

SDE cannot be negative. If the sum is below zero, BrokerLens sets SDE to zero.

MATERIAL LIMITATION

The MVP trusts the entered figures. It does not inspect a P&L;, tax return, general ledger, payroll file, or supporting invoice to confirm that an add-back is legitimate, recurring, or already counted elsewhere.

Demo SDE calculation

Item	Amount
Net profit	$188,000
Owner salary	+ $115,000
Interest	+ $12,000
Depreciation/amortization	+ $28,000
One-time add-backs	+ $22,000
Calculated SDE	$365,000

3. Starting multiple and adjustment rules

BrokerLens selects one fixed starting multiple from the chosen industry. These values are internal MVP assumptions and are not automatically refreshed from closed transaction databases.

Industry starting multiples

Industry	Starting multiple
Home services	3.10x
Professional services	3.40x
Restaurant and food	2.50x
Retail	2.55x
Manufacturing	3.70x
Other	2.90x

Revenue growth

Annual revenue growth	Multiple adjustment
10% or higher	+0.25x
5% to under 10%	+0.12x
0% to under 5%	0.00x
Below 0%	-0.30x

Recurring revenue

Recurring revenue share	Multiple adjustment
50% or higher	+0.25x
25% to under 50%	+0.12x
10% to under 25%	0.00x
Under 10%	-0.08x

3. Adjustment rules, continued

Owner dependence

Selection	Interpretation	Adjustment
Low	Team can operate with limited owner involvement.	+0.10x
Medium	Some owner knowledge or relationships must transfer.	-0.12x
High	Owner is essential to operations, delivery, or sales.	-0.35x

Largest customer concentration

Largest customer share	Adjustment
Under 15%	+0.08x
15% through 30%	0.00x
Over 30% through 50%	-0.18x
Over 50%	-0.30x

Lease stability

Years remaining	Adjustment
Under 2 years	-0.20x
2 to under 5 years	0.00x
5 years or more	+0.08x

Local market growth

Manually entered local growth	Adjustment
2% or higher	+0.15x
1% to under 2%	+0.06x
0% to under 1%	0.00x
Below 0%	-0.12x

LOCAL GROWTH IS NOT AUTOMATED

OpenAI may find local-growth evidence, but it does not currently update this field. The user must enter the percentage manually.

4. Final range, asking price, and confidence

Adjusted multiple

```
Adjusted multiple = starting multiple + all applicable quality adjustments
```

After all adjustments are added, the midpoint multiple is restricted to a minimum of 1.50x and a maximum of 5.50x.

Valuation range

```
Low multiple = max(1.25x, adjusted multiple - 0.40x)
Midpoint multiple = adjusted multiple
High multiple = adjusted multiple + 0.40x
```

```
Value = SDE x applicable multiple + inventory + excess assets - assumed debt
```

Each value is prevented from falling below zero and is rounded to the nearest $5,000.

Suggested asking price

```
Suggested asking price = high estimate x 1.06
```

The fixed 6% premium is intended to create negotiation room. It is not based on current buyer demand, days on market, or observed list-to-sale ratios.

Likely sale range

```
Likely sale low = low estimate x 1.02
Likely sale high = midpoint estimate x 1.03
```

These are fixed MVP assumptions rather than statistically trained predictions.

4. Confidence score

The displayed confidence percentage is a completion score. It is not a measured probability that the valuation is correct.

```
Confidence = 56 + (6 x number of completed required fields)
```

Completion check	Pass condition
Business name	Not blank
City	Not blank
Business description	Not blank
Net profit	Not equal to zero
Research domains	Not blank

The score is restricted to a minimum of 56% and maximum of 86%. Completing all five checks produces 86%.

WHAT THE SCORE DOES NOT MEASURE

It does not measure financial-statement quality, add-back validity, source quality, comparable-sale relevance, forecast reliability, economic volatility, or the accuracy of AI research.

5. Complete worked example

The preloaded demo represents Desert Air Mechanical, a Phoenix home-services business. The following reproduces the exact current calculation.

Step A - SDE

```
$188,000 + $115,000 + $12,000 + $28,000 + $22,000 = $365,000 SDE
```

Step B - adjusted multiple

Factor	Adjustment	Running multiple
Home-services starting multiple	3.10x	3.10x
11% revenue growth	+0.25x	3.35x
24% recurring revenue	0.00x	3.35x
Medium owner dependence	-0.12x	3.23x
7% largest customer share	+0.08x	3.31x
Four years on lease	0.00x	3.31x
1.8% local growth	+0.06x	3.37x

The adjustments produce a 3.37x midpoint multiple, with a 2.97x low multiple and 3.77x high multiple.

5. Complete worked example, continued

Step C - transaction adjustment

```
$62,000 inventory + $18,000 excess assets - $35,000 debt = +$45,000
```

Step D - valuation range

Output	Calculation	Rounded result
Low estimate	$365,000 x 2.97 + $45,000	$1,130,000
Midpoint estimate	$365,000 x 3.37 + $45,000	$1,275,000
High estimate	$365,000 x 3.77 + $45,000	$1,420,000
Suggested asking price	$1,420,000 x 1.06	$1,505,000
Likely sale low	$1,130,000 x 1.02	$1,155,000
Likely sale high	$1,275,000 x 1.03	$1,315,000

DEMO RESULT

Estimated market value: $1,130,000 to $1,420,000
Suggested asking price: $1,505,000
Displayed confidence: 86% completion score

6. OpenAI research workflow

The research feature calls the OpenAI Responses API only when OPENAI_API_KEY is configured. It performs web search inside user-approved domains and returns a concise market report with source links.

Business information sent to OpenAI

Sent to research	Not sent to research
Business name Industry City and state Description	Net profit SDE Owner salary Interest Depreciation Add-backs Recurring revenue Owner dependence Customer concentration Lease term Inventory Excess assets Debt Current valuation

Default approved sources

Domain	Intended evidence
census.gov	Population, income, establishment, and demographic data.
bls.gov	Employment, wages, hiring pressure, and labor-market data.
bea.gov	Regional economic output, income, and growth data.
bizbuysell.com	Business listings, transaction commentary, and valuation evidence.

The user may replace the defaults with up to 20 valid domains. Web search is restricted to the resulting domain list.

Research questions

- Local population, income, establishment, or demand growth.
- Industry employment, wages, and hiring pressure.
- Local competition or business-density signals.
- Small-business transaction or valuation evidence when available.

6. OpenAI research settings and output

Setting	Current value
Default model	gpt-5.6-sol, unless OPENAI_MODEL overrides it
Reasoning effort	Low
Web-search context	Medium
Tool choice	Automatic
Request storage	Disabled
Timeout	45 seconds
Returned findings	Up to five
Returned citation links	Up to eight
Vercel function maximum	60 seconds

Requested AI output

OpenAI is instructed to return a two-to-three-sentence broker summary and three to five signals. Each signal includes a short title, one specific finding, and a source domain. Available source annotations and consulted URLs are deduplicated and displayed as clickable citations.

Research safety controls

- Webpages are treated as untrusted evidence.
- The model is told to ignore instructions found inside source pages.
- The model is told not to invent figures.
- Facts should be separated from inference.
- The result is labeled supporting research, not a certified appraisal.

> **RESEARCH IS CURRENTLY DISCONNECTED FROM PRICE**
>
> The returned summary, signals, and citations are displayed in the Market section. They do not change SDE, the starting multiple, any adjustment, local growth, the valuation range, the asking price, the likely sale range, or the confidence score.

7. What is and is not used

Used in the dollar estimate

- Net profit, owner salary, interest, depreciation/amortization, and add-backs.
- The selected fixed industry starting multiple.
- Revenue growth, recurring revenue, owner dependence, customer concentration, lease term, and manually entered local growth.
- Inventory, excess assets, and buyer-assumed debt.
- Fixed range, negotiation-margin, likely-sale, floor, cap, and rounding rules.

Used only for AI research

- Business name, industry, location, description, and approved domains.

Used only for the completion score

- Name, city, description, net profit, and research domains.

Not currently used anywhere

- Confirmed closed-sale comparable records.
- Historical financial trends beyond the single entered growth percentage.
- Working-capital requirements, capital expenditures, taxes, financing terms, or buyer synergies.
- Lease transferability, market rent, renewal options, or landlord approval.
- Inventory age, condition, obsolescence, or liquidation value.
- Licenses, intellectual property, legal claims, environmental exposure, or key-person agreements.
- AI-generated numerical adjustments or AI-generated final value.

8. Limitations and recommended next stage

Current limitations

Limitation	Why it matters
Fixed industry multiples	They are not refreshed from actual comparable sales.
Unverified financial inputs	Incorrect add-backs or earnings can materially distort value.
Categorical adjustments	A value just above a threshold may change abruptly.
Manual local-growth input	The user may select a figure inconsistent with the research.
AI research is qualitative only	Useful evidence does not affect the calculated price.
Completion-based confidence	The percentage may appear more statistically meaningful than it is.
Dollar-for-dollar inventory treatment	Obsolete or already-included assets may be double counted.
No backtesting	The model has not been measured against known sale outcomes.

Recommended next-stage architecture

- Document extraction: parse uploaded P&Ls;, tax returns, payroll summaries, and spreadsheets into a reviewable financial schema.
- Source retrieval: gather current public economic data and transaction evidence from approved sources.
- Structured AI analysis: have OpenAI propose bounded, source-linked adjustments rather than an unrestricted final price.
- Broker approval: require the user to accept, reject, or modify each AI-proposed adjustment.
- Deterministic application: apply approved adjustments through the transparent valuation engine.
- Backtesting: compare estimated values with known transactions and tune the model based on measured error.

TARGET PRODUCT PRINCIPLE

BrokerLens should not be an AI that guesses a business value. It should be an evidence engine that gathers financial and market data, calculates a transparent valuation, and uses AI to connect sourced evidence to controlled adjustments.

Appendix: exact formula reference

Output	Exact current rule
SDE	max(0, profit + owner salary + interest + D&A; + add-backs)
Adjusted multiple	clamp(starting multiple + adjustment sum, 1.50x, 5.50x)
Low multiple	max(1.25x, adjusted multiple - 0.40x)
High multiple	adjusted multiple + 0.40x
Balance adjustment	inventory + excess assets - assumed debt
Low value	round to $5,000: max(0, SDE x low multiple + balance adjustment)
Midpoint value	round to $5,000: max(0, SDE x adjusted multiple + balance adjustment)
High value	round to $5,000: max(0, SDE x high multiple + balance adjustment)
Asking price	round to $5,000: high value x 1.06
Likely sale low	round to $5,000: low value x 1.02
Likely sale high	round to $5,000: midpoint value x 1.03
Confidence	clamp(46 + five points per completed check, 46, 86)

USE OF THIS DOCUMENT

This document describes the BrokerLens MVP as implemented on July 22, 2026. It is a technical-product methodology summary, not valuation advice, an appraisal standard, or a representation that any estimated price can be achieved in a sale.

Source of truth

Valuation engine: lib/valuation.ts
OpenAI research route: app/api/research/route.ts
Project: BrokerLens